Soldering Iron



Category: **Welding & Soldering** Type: **Full SOP** Priority: **High** 21 labs

Covers these lab items: Electric Soldering Iron; Soldering equipment; Soldering Iron; Soldering Iron (khadya); Soldering irons; Soldering kit; Soldering Kit with Accessories; Soldering Kits – Basic and Battery powered

What it is used for

A soldering iron heats solder to join suitable electrical components and wires. Correct temperature control, ventilation, and safe handling help prevent burns and poor connections.

Main parts

- Heated metal tip transfers heat to the joint.
- Heating element produces heat inside the iron.
- Insulated handle helps the user hold the tool safely.
- Power cord or battery supplies energy, depending on the model.
- Temperature control, if fitted, adjusts the tip temperature.
- Stand holds the hot iron when it is not being used.
- Sponge or brass wool helps clean the tip.

Before you start

1. Read the equipment instructions and confirm the iron suits the soldering task.
2. Wear safety goggles and work in a well-ventilated area with suitable fume extraction.
3. Inspect the cord, plug, handle, tip, stand, and temperature control for damage.
4. Place the stand on a stable, heat-resistant surface away from paper and plastics.
5. Prepare suitable solder, compatible flux, and the correct tip for the joint.
6. Secure the circuit or wire and switch off its power before soldering.

How to use

1. Place the soldering iron securely in its stand before connecting power.
2. Set the recommended temperature for the solder, tip, and components being used.
3. Switch on the iron and wait for it to reach working temperature.
4. Clean the tip using the recommended sponge or brass wool.
5. Apply a small amount of suitable solder to tin the clean tip when needed.
6. Heat the component lead and pad or wire joint together.
7. Feed solder into the heated joint rather than directly onto the tip.
8. Use only enough solder to form a neat, wetted connection.
9. Remove solder first, then lift the iron away and allow the joint to cool.
10. Switch off, unplug the iron, and leave it in the stand until fully cool.

⚠ Safety

- Never touch the metal tip or freshly soldered joint; both can cause serious burns.
- Keep the hot iron in its stand and never leave it unattended while powered.
- Use effective ventilation or fume extraction and avoid breathing soldering fumes.
- Keep the tool away from water, flammable materials, and trailing cables.
- Do not solder an energised circuit or a battery connected to the circuit.
- Wash hands after handling solder and flux, especially before eating.

Care & storage

- Switch off and unplug the iron before cleaning or replacing its tip.
- Clean and tin the tip as recommended to reduce oxidation.
- Do not file or scrape plated tips unless the manufacturer specifically permits it.
- Store the fully cooled iron, solder, and accessories in a dry, secure place.

Common mistakes to avoid

- Touching the hot tip to check whether the iron is ready.
- Heating the solder instead of heating the joint surfaces properly.
- Using excessive solder or keeping the iron on components too long.
- Leaving the iron powered on or resting it outside its stand.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

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